Professional Online Academy Vu Topper RM

MTH603-Numerical Analysis Update MCQ'S Final Term By Vu Topper RM

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Question No:1 **(Marks:1)** **Vu-Topper RM**

Given that with the initial condition Using Modified Euler's method, for the range , h = 0.1 is

A. 1.2
B. 1.1
C. 2.1
D. 1.0

Question No:2 **(Marks:1)** **Vu-Topper RM**

Which of the following is the Global Error for Simpson's 3/8 Rule while integrating 'f(x) = Cosx' in the interval of [0,pi] of equally spaced subinterval of width 'h =pi/6' and intermediate point x = pi/2?

A. -pi/80
B. 0
C. pi/80
D. 1

Question No:3 **(Marks:1)** **Vu-Topper RM**

If , and h= 0.2. Then, of Runge-Kutta method is:

A. 0.1
B. 0.4
C. 0.2
D. 0.3

Question No:4 **(Marks:1)** **Vu-Topper RM**

The single definite integral of a function is called ……

A. Volume of the curve
B. Area under the curve
C. Length of the curve
D. None of the given choices

Question No:5 **(Marks:1)** **Vu-Topper RM**

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Given that with the initial condition find the 2nd term in Taylor series when t=1, y/ =1.2, and h=0.1.

A. 1.2
B. 1.4
C. 1.5
D. 1.3

Question No:6 **(Marks:1)** **Vu-Topper RM**

In solving the differential equation
By Euler's method y(0.2) is calculated as

A. 0.2
B. 1.01
C. 1.02
D. 0.1

Question No:7 **(Marks:1)** **Vu-Topper RM**

Given that with the initial condition and h= o.1. Then, of Runge-Kutta method is:

A. 0.3
B. 0.4
C. 0.1
D. 0.2

Question No:8 **(Marks:1)** **Vu-Topper RM**

Given that with the initial condition find the 3rd term in Taylor series when t=1.5 and y// =0.6

A. 1.45
B. 1.35
C. 1.25
D. 1.15

Question No:9 **(Marks:1)** **Vu-Topper RM**

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Given that with the initial condition find the 2nd term in Taylor series when t=1.5 and y/ =1.2.

A. 0.5
B. 0.4
C. 0.6
D. 0.7

Question No:10 (Marks:1) Vu-Topper RM

The step size "h" in numerical integration over the interval [a,b] is defined as

A. h=(b-a)/n
B. h=(b+a)/n
C. h=(a-b)/n
D. h=(b/a)/n

Question No:11 (Marks:1) Vu-Topper RM

While using the Composite Trapezoidal form for integrating y = f(x) in [0,10] which is subdivided in equally spaced interval of width 'h =2', then which of the following is the area of associated trapezoidal strip over subinterval:[2,4] ?

A. (y2 + y4)/2
B. (y2 – y4)/2
C. (y2 – y4)
D. (y2 + y4)

Question No:12 (Marks:1) Vu-Topper RM

Given that with the initial condition find the 2nd term in Taylor series when t=0.2 and y/= 0.3.

A. 0.06
B. 0.01
C. 0
D. 0.02

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Question No:13 **(Marks:1)** **Vu-Topper RM**

Given that with the initial condition find the 3rd term in Taylor series when t=1.5 and y/ =1 and y// =1.2

A. 0.1
B. 0.2
C. 0.3
D. 0.4

Question No:14 **(Marks:1)** **Vu-Topper RM**

Given that with the initial condition find the 3rd term in Taylor series when t=1.5 and y// =0.6

A. 1.35
B. 1.45
C. 1.15
D. 1.25

Question No:15 **(Marks:1)** **Vu-Topper RM**

Given that with the initial condition find the 2nd term of Taylor series when t=0.5 and y/ =0.

A. 0
B. 0.3
C. 0.1
D. 0

Question No:16 **(Marks:1)** **Vu-Topper RM**

---------- difference is a symmetric function of its arguments.

A. Forward
B. Backward
C. None
D. Divided

Question No:17 **(Marks:1)** **Vu-Topper RM**

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What will be the value of first order divided difference f[1,5]for the following data x:0 1 5 y:2 1 5

A. 1
B. 3
C. 2
D. 0

Question No:18 **(Marks:1)** **Vu-Topper RM**

Given the following data x:0 1 4 8 y:1 1 8 16 Value of 1st order divided difference f[4,8] is

A. 8
B. 6
C. 2
D. 4

Question No:19 **(Marks:1)** **Vu-Topper RM**

If y(x) is approximated by a polynomial of degree n then the error is given by Pn(x)

A. $\varepsilon(x) = y(x) \div Pn(x)$
B. $\varepsilon(x) = y(x) + Pn(x)$
C. $\varepsilon(x) = y(x) \times Pn(x)$
D. $\varepsilon(x) = y(x) - Pn(x)$

Question No:20 **(Marks:1)** **Vu-Topper RM**

In Lagrange's interpolation, for n values of y corresponding to n values of x, we can represent the function f (x) by a polynomial of degree

A. n
B. n-1
C. n+1
D. n+2

Question No:21 **(Marks:1)** **Vu-Topper RM**

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Given the following data x:1 2 5 y:1 4 10 Value of 1st order divided difference f[2 , 5] is

A. 0
B. 1
C. 2
D. -2

Question No:22 **(Marks:1)** **Vu-Topper RM**

If there are (n+2) values of y corresponding to (n+2) values of x, then we can represent the function f(x) by a polynomial of degree

A. n-1
B. n+1
C. n+2
D. n

Question No:23 **(Marks:1)** **Vu-Topper RM**

Given the following data x:1 2 7 11 y:6 10 13 37 Which formula is useful in finding the interpolating polynomial?

A. Newton's backward difference interpolation formula
B. Lagrange's interpolation formula
C. Newton's forward difference interpolation formula
D. None

Question No:24 **(Marks:1)** **Vu-Topper RM**

Given the following data x:4 5 7 10 y:46 102 294 346 Value of 1st order divided difference f[5 , 7] is

A. 94
B. 96
C. 92
D. 91

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Question No:25 **(Marks:1)** **Vu-Topper RM**

Lagrange's interpolation formula is used when the values of the independent variable are

A. Constant

B. Not equally spaced

C. None

D. Equally spaced

Question No:26 **(Marks:1)** **Vu-Topper RM**

Given the following data x:0 1 4 y:2 1 4 Value of first order divided difference y[0,1]is

A. -1

B. -2

C. 1

D. 2

Question No:0_1 **(Marks:1)** **Vu-Topper RM**

At which of the following point the derivatives or slopes the functions $f(x) = x - 2$ and $g(x) = x + 2$ may differ?

A. differ for every value of 'x'

B. does not differ for any value of 'x'

C. x = 2

D. x = -2

Question No:27 **(Marks:01)** **Vu-Topper RM**

In Simpson's rule, we assume that f(x) is continuous on [a, b] and we divide [a, b] into an …… number n of subintervals of equal length.

A. Odd

B. Even **ok**

C. Prime

D. None of the given choices

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Question No:28 **(Marks:01)** **Vu-Topper RM**

Geometrically, in Simpson's 1/3 Rule, we find the area of -------- strip/strips at time under a curve of given function.

A. Three **ok**

B. five

C. one

D. two

Question No:29 **(Marks:01)** **Vu-Topper RM**

To improve the accuracy of the derivative of the function, which of the following method is more helpful?

A. Extrapolation **Page 1**

B. Interpolation

C. Divided difference

D. Central difference

Question No:30 **(Marks:01)** **Vu-Topper RM**

To take the derivative of f(x) = 2x in the interval [-3,3], which of the following partition of subintervals will be suitable?

A. Equally spaced

B. Unequally spaced

C. Union of equally spaced and unequally spaced intervals.

D. Any arbitrary partition will work

Question No:31 **(Marks:01)** **Vu-Topper RM**

In Composite Trapezoidal formula for integrating a Tabular function, we can approximate it with a polynomial whose ---------- order derivative vanishes.

A. Third

B. First

C. Second **ok**

D. Fourth

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Question No:32 **(Marks:01)** **Vu-Topper RM**

If the area under 'f(x) = x' in interval [0,2] is subdivided into two equal sub-intervals of width '1' with left end points, then which of the following will be the Truncation Error provided that I(definite integral) = 2 and approximate sum = 3 ?

A. 0
B. 1 **ok**
C. -1
D. 3

Question No:33 **(Marks:01)** **Vu-Topper RM**

The percentage error in numerical integration is defined as

A. = (Theoretical Value-Experiment Value)* Experiment Value*100
B. = (Theoretical Value +Experiment Value)/ Experiment Value*100
C. = (Theoretical Value-Experiment Value)/ Theoretical Value *100
D. = (Theoretical Value-Experiment Value)/ Experiment Value*100

Question No:34 **(Marks:01)** **Vu-Topper RM**

Two-point formula for the first derivative is defined as

A. Y(xi+H)-y(xi-h)/2h
B. Y(xi+H)-y(xi-h)/2h
C. Y(xi+H)-y(xi-h)/2^h
D. Y(xi+H)+y(xi-h)/2^h

Question No:35 **(Marks:01)** **Vu-Topper RM**

Trapezoidal rule of integration of a definite integral is of?

A. o(h2)
B. o(h3)
C. o(h4)
D. none of the given choice

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Question No:36 **(Marks:01)** **Vu-Topper RM**

To evaluate numerically a double integral over a rectangular region bounded by the lines x=a, x=b, y=c, y=d we shall employ either trapezoidal rule of Simpson's rule, repeatedly with respect to ||| variable at a time

A. One
B. Two
C. Three
D. None of given option

Question No:37 **(Marks:01)** **Vu-Topper RM**

Simpson's rule is a numerical method that approximates the value of definite integral by using …. Polynomials

A. Linear
B. **Quadratic polynomial**
C. Cubic
D. None of the given choice

Question No:38 **(Marks:01)** **Vu-Topper RM**

In Simpson's rule we assume that f(x) is continuous on [a,b] and we divide [a,b] into an number n of subintervals of equal length.

A. odd
B. Even
C. Prime
D. None of given option

Question No:39 **(Marks:01)** **Vu-Topper RM**

The step size |h| in numerical integration over the interval [a, b] is defined as

A. h= [b-a]/n
B. h= [b+a]/n
C. h= [a-b]/n
D. h= [b/a]/n

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Question No:40 **(Marks:01)** **Vu-Topper RM**

Which of the following is the Global Error for Simpson's 3/8 Rule while integrating 'f(x) = Cosx' in the interval of (0, pi) of equally spaced subinterval of width h'= 6' and intermediate point x = pi/2?

A. -pi / 80
B. Pi/80
C. 0
D. 1

Question No:41 **(Marks:01)** **Vu-Topper RM**

Eigenvalues of a symmetric matrix are all _____ .

A. Real
B. complex
C. zero
D. positive

Question No:42 **(Marks:01)** **Vu-Topper RM**

Below are all the finite difference methods EXCEPT _____.

A. jacobi's method
B. newton's backward difference method
C. Stirlling formula
D. Forward difference method

Question No:43 **(Marks:01)** **Vu-Topper RM**

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix. where symbol ^ shows power.

A. (x-1)^3
B. (x+1)^3
C. x^3-1
D. x^3+1

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Question No:44 **(Marks:01)** **Vu-Topper RM**

Eigenvalues of a symmetric matrix are all _____.

Select correct option:

A. real

B. zero

C. positive

D. negative

Question No:45 **(Marks:01)** **Vu-Topper RM**

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix. where symbol ^ shows power.

A. (x-1)^3

B. (x+1)^3

C. x^3-1

D. x^3+1

Question No:46 **(Marks:01)** **Vu-Topper RM**

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its _____.

Select correct option:

A. main diagonal

B. last column

C. last row

D. first row

Question No:47 **(Marks:01)** **Vu-Topper RM**

Richardson extrapolation method is used to improve the rate of convergence of a ……

A. Sequence

B. Series

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Question No:48 **(Marks:01)** **Vu-Topper RM**

Simpson's 3/8 rule represents the area between the curve y = f(x) in the interval say [a,b] above x-axis by approximating the given curve by the ----------.

A. Cubic curve through one point
B. Cubic curve through two points
C. Cubic curve through three points ok
D. Cubic curve through four points

Question No:49 **(Marks:01)** **Vu-Topper RM**

Three-point formula for the second derivative is defined as

A. y(xi+h)+2y(xi)+y(xi-h)/h^2
B. y(xi+h)-2y(xi)+y(xi-h)/h^2
C. y(xi+h)-2y(xi)-y(xi-h)/h^2
D. y(xi+h)-2y(xi)+y(xi-h)/2h

Question No:50 **(Marks:01)** **Vu-Topper RM**

To evaluate a definite integral of tabular function f(x), piecewise linear approximation led to ---------.

A. Trapezoidal Method ok
B. Simpson's 1/3 Rule
C. Simpson's 3/8 Rule
D. Romberg's Method

Question No:51 **(Marks:01)** **Vu-Topper RM**

Which of the following is the Global Error for Trapezoidal's Rule while integrating f(x) = x^2 in the interval of [0,2] of equally spaced subinterval of width h =1 and intermediate point x=1?

A. 1/3
B. -1/3
C. -1/2
D. 1/2 ok

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Question No:52 **(Marks:01)** **Vu-Topper RM**

The determinant of a_matrix is the product of the diagonal elements.

A. Diagonal
B. Upper triangular
C. Lower triangular
D. Scalar

Question No:53 **(Marks:01)** **Vu-Topper RM**

Power method is applicable if the eigen values are _____.

A. real and distinct
B. real and equal
C. positive and distinct
D. negative and distinct

Question No:54 **(Marks:01)** **Vu-Topper RM**

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its _____.

A. main diagonal
B. last column
C. last row
D. first row

Question No:55 **(Marks:01)** **Vu-Topper RM**

Eigen values of a_____ matrix are all real.

A. Symmetric
B. Antisymmetric
C. Rectangular
D. Triangular

Question No:56 **(Marks:01)** **Vu-Topper RM**

Simpson's rule is a numerical method that approximates the value of a definite integral by using polynomials.

A. Quadratic

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B. Linear
C. Cubic
D. Quartic

Question No:57 (Marks:01) Vu-Topper RM
The Trapezoidal rule is a numerical method that approximates the value of a_____.
A. Indefinite integral
B. Definite integral
C. Improper integral
D. Function

Question No:58 (Marks:01) Vu-Topper RM
The Trapezoidal Rule is an improvement over using rectangles because we have much less "missing" from our calculations. We used _____ to model the curve in trapezoidal Rule.
A. straight lines
B. curves
C. parabolas
D. constant

Question No:59 (Marks:01) Vu-Topper RM
Euler's Method numerically computes the approximate _____ of a function.
A. Antiderivative
B. Derivative
C. Error
D. Value

Question No:60 (Marks:01) Vu-Topper RM
If we wanted to find the value of a definite integral with an infinite limit, we can instead replace the infinite limit with a variable, and then take the limit as this variable goes to _____.

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A. Constant
B. Finite
C. Infinity
D. Zero

Question No:61 **(Marks:01)** **Vu-Topper RM**

Let A be an n ×n matrix. The number x is an eigenvalue of A if there exists a nonzero vector v such that _____.

A. Av = xv
B. Ax = xv
C. Av + xv=0
D. Av = Ax

Question No:62 **(Marks:01)** **Vu-Topper RM**

In Jacobi's Method, the rate of convergence is quite _____ compared with other
methods.

A. Slow
B. Fast
C. Both
D. None

Question No:63 **(Marks:01)** **Vu-Topper RM**

Numerical solution of 2/3 up to four decimal places is _____.

A. 0.667
B. 0.6666
C. 0.6667
D. 0.66667

Question No:64 **(Marks:01)** **Vu-Topper RM**

The first langrange polynomial with equally spaced nodes produced the formula for _____.

A. Simpson's rule

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B. Trapezoidal rule
C. Newton's method
D. Richardson's method

Question No:65 **(Marks:01)** **Vu-Topper RM**
The Euler method is numerically unstable because of _____ convergence of error.

A. Slow
B. Fast
C. Moderate
D. No

Question No:66 **(Marks:01)** **Vu-Topper RM**
Generally, Adams methods are superior if output at _____ points is needed.

A. Many
B. Two
C. Single
D. At most

Question No:67 **(Marks:01)** **Vu-Topper RM**
Symbol used for forward differences is

A. ∇
B. Δ
C. δ
D. μ

Question No:68 **(Marks:01)** **Vu-Topper RM**
The relationship between central difference operator and the shift operator is given by

A. $\delta = E - E^{-1}$
B. $\delta = E + E^{-1}$
C. $\delta = E1/2 + E-1/2$

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D. δ =E1/2-E-1/2

Question No:69 **(Marks:01)** **Vu-Topper RM**

Muller's method requires --------starting points

A. 1
B. 2
C. 3
D. 4

Question No:70 **(Marks:01)** **Vu-Topper RM**

If we retain r+1 terms in Newton's forward difference formula, we obtain a polynomial of degree ---- agreeing with $y_x$at x_0, x_1, …. X_n.

A. r+2
B. r+1
C. **r**
D. r-1

Question No:71 **(Marks:01)** **Vu-Topper RM**

Octal number system has the base ----------

A. 2
B. 8
C. 10
D. 16

Question No:72 **(Marks:01)** **Vu-Topper RM**

Newton's divided difference interpolation formula is used when the values of the are

A. Equally spaced
B. Not equally spaced
C. Constant
D. None of the above

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Question No:73 **(Marks:01)** **Vu-Topper RM**

Rate of change of any quantity with respect to another can be modeled by

A. An ordinary differential equation
B. A partial differential equation
C. A polynomial equation
D. None of the given choices

Question No:74 **(Marks:01)** **Vu-Topper RM**

Adam-Moulton P-C method is derived by employing

A. Newton's backward difference interpolation formula
B. Newton's forward difference interpolation formula
C. Newton's divided difference interpolation formula
D. None of the given choices

Question No:75 **(Marks:01)** **Vu-Topper RM**

Bisection method is …….. method

A. Bracketing Method
B. Open
C. Random
D. none

Question No:76 **(Marks:01)** **Vu-Topper RM**

Newton Raphson method is …….. method

A. Bracketing Method
B. Open
C. Random
D. none

Question No:77 **(Marks:01)** **Vu-Topper RM**

Bisection and false position methods are also known as

A. bracketing method
B. open method

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C. random
D. None of these

Question No:78 **(Marks:01)** **Vu-Topper RM**

The Inverse of a matrix can only be found if the matrix is Singular

A. Non singular
B. Scalar
C. None
D. Diagonal

Question No:79 **(Marks:01)** **Vu-Topper RM**

If f (x) contains trigonometric, exponential or logarithmic functions then this equation is known as

A. Transcendental equation
B. Algebraic
C. Polynomial
D. Linear

Question No:80 **(Marks:01)** **Vu-Topper RM**

In interpolation δ is used to represent the

A. Forward difference
B. Central difference
C. Backward difference
D. None of these

Question No:81 **(Marks:01)** **Vu-Topper RM**

The base of the decimal system is _____

A. 10
B. 0
C. 2
D. 8

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Question No:82 **(Marks:01)** **Vu-Topper RM**

P in Newton's forward difference formula is defined as

A. P=(x-x0)/h

B. P=(x+x0/h

C. $P=(x+x_n)/h$

D. $P=(x-x_n)/h$

Question No:83 **(Marks:01)** **Vu-Topper RM**

Newton's divided difference interpolation formula is used when the values of the are

A. Equally spaced

B. Not equally spaced

C. Constant

D. None of the above

Question No:84 **(Marks:01)** **Vu-Topper RM**

Given the following data

X	0	1	2	4
F(x)	1	1	2	5

The value of f(2,4) is

A. 1.5

B. 3

C. 2

D. 1

Question No:85 **(Marks:01)** **Vu-Topper RM**

To evaluate a definite integral of tabular function f(x), piecewise quardratic approximation led to ---------

A. Simpson's Rule

B. Trapezoidal Method **ok**

Question No:86 **(Marks:01)** **Vu-Topper RM**

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Vu Topper RM

At which of the following points the Minimum value of 2nd derivative of function f(x) = -(2/x) in the interval:[1,4] exits?

A. At x=1 **ok**

B. At x=2

C. At x=3

D. At x=4

Question No:87 **(Marks:01)** **Vu-Topper RM**

In double integration, we keep one variable say x fixed and ……

A. Reliable the other variable y

B. Varying the other variable y **ok**

Question No:88 **(Marks:01)** **Vu-Topper RM**

In Simpson's 3/8 rule, the global error is of ……

Select correct option:

A. O(h2)

B. O(h3)

C. O(h4) **ok**

D. None of the given choices

Question No:89 **(Marks:01)** **Vu-Topper RM**

Multistep method does not improves the accuracy of the answer at each step.

True

False

Question No:90 **(Marks:01)** **Vu-Topper RM**

If the area under 'f(x) = x' in interval [0,2] is subdivided into two equal sub-intervals of width '1' with right end points, then which of the following will be the Truncation Error provided that I(definite integral) = 2 and approximate sum = 3 ?

A. 0 **ok**

B. 1

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C. -1
D. 2

Question No:91 **(Marks:01)** **Vu-Topper RM**

yn+1=yn+16(K1+2K2+2k3+k4) is known as:

A. Taylor's formula
B. Simpson's formula
C. Euler's formula
D. Runge-Kutta's formula **ok**

Question No:92 **(Marks:01)** **Vu-Topper RM**

Given that dydt=t+y√ with the initial condition y0=1att0=0 find the 2nd term in Taylor series when t=1, y/ =0.2, and h=0.1.

A. 0
B. 0.1
C. 0.2
D. 0.3 **ok**

Question No:93 **(Marks:01)** **Vu-Topper RM**

Given that dydt=y−ty+t with the initial condition y=1,t=0 Using Euler's method, y ath=0.01; the value of y(0.01)is

A. 1.1
B. 1.2
C. 1.02
D. 1.03

Question No:94 **(Marks:01)** **Vu-Topper RM**

In Trapezoidal rule, we assume that f(x) is continuous on [a, b] and we divide [a, b] into n subintervals of equal length using the ……points.

A. N **ok**
B. N+1
C. n-1
D. None of the given choices

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Question No:95 **(Marks:01)** **Vu-Topper RM**

To evaluate numerically a double integral over a rectangular region bounded by the lines x = a, x =b, y = c, y = d we shall employ either trapezoidal rule or Simpson's rule, repeatedly with respect to ……variable at a time.

A. one
B. two ok
C. three
D. None of the given choice

Question No:96 **(Marks:01)** **Vu-Topper RM**

The double definite integral of a function is called……

A. Volume under the curve ok
B. Length of the curve
C. Area under the curve
D. None of the given choices

Question No:97 **(Marks:01)** **Vu-Topper RM**

If the given tabular function f(x) is approximated by the polynomial 'P1(x) = x+1' then which of the following polynomial will approximate the derivative of f(x) ?

Select correct option:

A. x
B. x-1
C. 2x
D. 1

Question No:98 **(Marks:01)** **Vu-Topper RM**

Geometrically the definite integral of any continuous function f(x) in the interval [a,b] gives ----------.

A. Area under f(x) on [a,b] ok
B. Area of Trapezoid with dimension of 'a' and 'b'

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C. Volume with dimensions f(x), ‘a’ and ‘b’
D. Length of segment AB on real line

Question No:99 **(Marks:01)** **Vu-Topper RM**
In Simpson’s rule, we can estimate the integral by ……the areas under the parabolic arcs through three successive points.
A. None of the given choices
B. Multiplying ok
C. Adding
D. Subtracting

Question No:100 **(Marks:01)** **Vu-Topper RM**
Integration is a ……process.
A. Subtracting
B. Summing ok
C. Dividing
D. None of the given choices

Question No:101 **(Marks:01)** **Vu-Topper RM**
In Simpson’s 1/3 rule, we divide the interval of integration [a, b] into an ……number of sub-intervals.
A. Odd
B. Prime
C. Even
D. None of the given choices

Question No:102 **(Marks:01)** **Vu-Topper RM**
The idea behind the predictor-corrector methods is to use a suitable combination of an explicit and an implicit technique to obtain a method with better convergence characteristics.
True
False

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Question No:103 **(Marks:01)** **Vu-Topper RM**
An improper integral is the limit of a definite integral as an endpoint of the interval of sinteg ration approaches either a specified real number or ∞ or - ∞ or, in some cases, as both endpoints approach limits.
True
False

Question No:104 **(Marks:01)** **Vu-Topper RM**
An indefinite integral may _____ in the sense that the limit defining it may not exist.
Diverge
Converge

Question No:105 **(Marks:01)** **Vu-Topper RM**
In Runge –Kutta Method, we do not need to calculate higher order derivatives and find greater accuracy.
True
False

Question No:106 **(Marks:01)** **Vu-Topper RM**
The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose antiderivative is not easy to obtain.
Antiderivative
Derivatives

Question No:107 **(Marks:01)** **Vu-Topper RM**
Generally, Adams methods are superior if output at many points is needed.
True
False

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Question No:108 **(Marks:01)** **Vu-Topper RM**
In Trapezoidal rule, the integral is computed on each of the sub-intervals by using linear interpolating formula, ie. For n=1 and then summing them up to obtain the desired integral.
True
False

Question No:109 **(Marks:01)** **Vu-Topper RM**
The predictor-corrector method an implicit method. (multi-step methods)
True
False

Question No:110 **(Marks:01)** **Vu-Topper RM**
In Simpson's Rule, we use parabolas to approximating each part of the curve. This proves to be very efficient as compared toTrapezoidal rule.
True
False

Question No:111 **(Marks:01)** **Vu-Topper RM**
The dominant or principal eigenvector of a matrix is an eigenvector corresponding to the
Eigen value of largest magnitude (for real numbers, largest absolute value) of that matrix,
True
False

Question No:112 **(Marks:01)** **Vu-Topper RM**
Central Difference method is the finite difference method.
True
False

Question No:113 **(Marks:01)** **Vu-Topper RM**
Iterative algorithms can be more rapid than direct methods.

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True
False

Question No:114 **(Marks:01)** **Vu-Topper RM**
Central difference method seems to be giving a better approximation, however it requires more computations.
True
False

Question No:115 **(Marks:01)** **Vu-Topper RM**
If x is an eigenvalue corresponding to eigenvalue of V of a matrix A. If a is any constant, then x – a is an eigen value corresponding to eigen vector V is an of the matrix A - a I.
True
False

Question No:116 **(Marks:01)** **Vu-Topper RM**
For differences methods we require the set of values
True
False

Question No:117 **(Marks:01)** **Vu-Topper RM**
The Power method can be used only to find the eigenvalue of A that is largest in absolute value—we call this eigenvalue the dominant eigenvalue of A.
True
False

Question No:118 **(Marks:01)** **Vu-Topper RM**
The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric definite matrices A.
Positive
Negative

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Question No:119 **(Marks:01)** **Vu-Topper RM**
The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric positive definite matrices A.
True
False

Question No:120 **(Marks:01)** **Vu-Topper RM**
An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to zero.
True
False

Question No:121 **(Marks:01)** **Vu-Topper RM**
An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to _____.
Unity
Zero

Question No:122 **(Marks:01)** **Vu-Topper RM**
The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its main diagonal.
(Bronshtein and Semendyayev 1997, p. 892).
True
False

Question No:123 **(Marks:01)** **Vu-Topper RM**
The Jacobi's method is a method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal. (Bronshtein and Semendyayev 1997, p. 892)
No
At least one

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Question No:124 **(Marks:01)** **Vu-Topper RM**
If n x n matrices A and B are similar, then they have the same eigenvalues (with the same multiplicities).
True
False

Question No:125 **(Marks:01)** **Vu-Topper RM**
If n x n matrices A and B are similar, then they have the _____ eigenvalues (with the same multiplicities).
Same
Different

Question No:126 **(Marks:01)** **Vu-Topper RM**
A 3 x 3 identity matrix have three and _____Eigen values.
Same
Different

Question No:127 **(Marks:01)** **Vu-Topper RM**
A 3 x 3 identity matrix have three and different eigen values.
True
False

Question No:128 **(Marks:01)** **Vu-Topper RM**
Power method is applicable if the Eigen values are real and distinct.
True
False

Question No:129 **(Marks:01)** **Vu-Topper RM**
Power method is applicable if the Eigen vectors corresponding the Eigen values are linearly
Independent **Page 6**
Dependent

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Question No:130 **(Marks:01)** **Vu-Topper RM**
The determinant of a diagonal matrix is the product of the diagonal elements.
True
False

Question No:131 **(Marks:01)** **Vu-Topper RM**
If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix , the eigenvalues of A are the diagonal entries of A.
True
False

Question No:132 **(Marks:01)** **Vu-Topper RM**
The Power method can be used only to find the eigenvalue of A that is largest in absolute value—we call this eigenvalue the dominant eigenvalue of A.
True
False

Question No:133 **(Marks:01)** **Vu-Topper RM**
Exact solution of 2/3 is not exists.
True
False

Question No:134 **(Marks:01)** **Vu-Topper RM**
The Jacobi's method is
a method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal.
No
Atleast one

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Question No:135 **(Marks:01)** **Vu-Topper RM**

While employing trapezoidal and Simpson rules to evaluate the double integral numerically, by using trapezoidal and Simpson rule over …….

Plane region

Real line

Question No:136 **(Marks:01)** **Vu-Topper RM**

1st ordered divided difference formula is defined as

y[x0,x1]=(y1+y0)/(x1-x0)

Question No:137 **(Marks:01)** **Vu-Topper RM**

At which of the following points the Maximum value of 2nd derivative of function f(x) = -(2/x) in the interval:[1,4] exits?

A. At x=2 **ok**

B. At x=3

C. At x=1

D. At x=4

Question No:138 **(Marks:01)** **Vu-Topper RM**

Bisection and false position methods are also known as bracketing method and are always

Divergent

Convergent

Question No:139 **(Marks:01)** **Vu-Topper RM**

Adams – Bashforth is a multistep method.

True

False

Question No:140 **(Marks:01)** **Vu-Topper RM**

Multistep method does not improve the accuracy of the answer at each step.

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False
True

Question No:141 **(Marks:01)** **Vu-Topper RM**
The need of numerical integration arises for evaluating the indefinite integral of a function that has no explicit antiderivative or whose antiderivative is not easy to obtain.
True
False

Question No:142 **(Marks:01)** **Vu-Topper RM**
Euler's method is only useful for a few steps and small step sizes; however Euler's method together with Richardson extrapolation may be used to increase the _____.
Order and accuracy
Divergence

Question No:143 **(Marks:01)** **Vu-Topper RM**
The absolute value of a determinant (|detA|) is the product of the absolute values of the eigenvalues of matrix A
True
False

Question No:144 **(Marks:01)** **Vu-Topper RM**
Eigenvectors of a symmetric matrix are orthogonal, but only for distinct eigenvalues.
True
False

Question No:145 **(Marks:01)** **Vu-Topper RM**
The central difference method is finite difference method.
True
False

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Question No:146 **(Marks:01)** **Vu-Topper RM**
The dominant eigenvector of a matrix is an eigenvector corresponding to the eigenvalue of largest magnitude (for real numbers, smallest absolute value) of that matrix.
True
False

Question No:147 **(Marks:01)** **Vu-Topper RM**
Direct method can more rapid than iterative algorithms
True
False

Question No:148 **(Marks:01)** **Vu-Topper RM**
By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.
True
False

Question No:149 **(Marks:01)** **Vu-Topper RM**
Differences methods find the _____ solution of the system.
Numerical
Analytical

Question No:150 **(Marks:01)** **Vu-Topper RM**
Two matrices with the same characteristic polynomial need not be similar.
True
False

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Question No:151 **(Marks:01)** **Vu-Topper RM**
If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix, the eigenvalues of A are the diagonal entries of A.
True
False

Question No:152 **(Marks:01)** **Vu-Topper RM**
The Jacobi iteration _____, if A is strictly diagonally dominant.
Converges
Diverges

Question No:153 **(Marks:01)** **Vu-Topper RM**
The Jacobi iteration converges, if A is strictly diagonally dominant.
True
False

Question No:154 **(Marks:01)** **Vu-Topper RM**
Exact solution of 2/3 is not exists.
True
False

Question No:155 **(Marks:01)** **Vu-Topper RM**
Euler's Method numerically computes the approximate derivative of a function.
True
False

Question No:156 **(Marks:01)** **Vu-Topper RM**
While employing Trapezoidal and Simpson Rules to evaluate the double integral numerically, by using Trapezoidal and Simpson rule over --------.
A. Plane region
B. Both

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C. None
D. Real line

Question No:157 (Marks:01) Vu-Topper RM
While employing Trapezoidal and Simpson Rules to evaluate the double integral numerically, by using Trapezoidal and Simpson rule with respect to -------- variable/variables at time
A. **Single ok**
B. Both
C. None
D. Double

Question No:158 (Marks:01) Vu-Topper RM
Which of the following method is simplest one to integrate numerically a given tabular function but give more error?
A. Simpson's 3/8 Rule
B. Simpson's 1/3 Rule
C. Trapezoidal method
D. Rectangular method

Question No:159 (Marks:01) Vu-Topper RM
Simpson's 3/8 rule is based on fitting …… points by a cubic.
A. Two
B. Three
C. Four Page 169 ok
D. None of the given choices

Question No:160 (Marks:01) Vu-Topper RM
We can improve the accuracy of trapezoidal and Simpson's rules using ……
A. Simpson's 1/3 rule

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B. Simpson's 3/8 rule
C. Richardson's extrapolation method Page 178 ok
D. None of the given choices

Question No:161 (Marks:01) Vu-Topper RM
Which of the following is the Cote's number (weighting coefficient) for the function: f(x) = x+1 in the interval [0,1]?
A. 3/2
B. -3/2
C. ½ ok
D. -1/2

Question No:162 (Marks:01) Vu-Topper RM
The area of a trapezoid is obtained by adding the area of a ……and a triangle.
A. Rectangle ok
B. Square
C. Circle
D. None of given

Question No:163 (Marks:01) Vu-Topper RM
Which of the following reason(s) lead towards the numerical integration methods?
A. All above choices are true
B. Integrand is given in tabular form ok
C. Analytical evaluation of integral is impossible
D. Analytical evaluation of integral is very complicated

Question No:164 (Marks:01) Vu-Topper RM
In Simpson's 3/8 rule, we divide the interval of integration into n sub-intervals. Where n is divisible by.....
A. 2

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B. 3
C. 4 ok
D. None of the given choices

Question No:165 (Marks:01) Vu-Topper RM
For a function 'f(x) = x', with a step size of 'h=0.01', which of the following gives the 1st derivative at x =1 by using two point formula?
A. Y'=(x)=1+Some truncation Error
B. Y'=(x)=1.01+Some truncation Error
C. Y'=(x)=.0.1+Some truncation Error
D. Y'=(x)=0.1+Some truncation Error

Question No:166 (Marks:01) Vu-Topper RM
Simpson's 1/3 rule is based on fitting three points with a ……
A. Cubic
B. **Quadratic Page 5 ok**

Question No:167 (Marks:01) Vu-Topper RM
In Newton-Cotes formula for finding the definite integral of a tabular function, which of the following is taken as an approximate function then find the desired integral?
A. Polynomial Function Page 166
B. Exponential Function
C. Logarithmic Function
D. Trigonometric Function

Question No:168 (Marks:01) Vu-Topper RM
Trapezoidal rule of integration of a definite integral is of……
A. O(h2) ok
B. O(h3)
C. O(h4)
D. None of the given choices

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Question No:169 **(Marks:01)** **Vu-Topper RM**

While using the composite trapezoidal form for integrating y=f(x) in [0,10] which is subdivided in equally space interval of width h=2 then which of the following is the area of associated trapezoidal strip over subinterval: [2,4]?

A. (y2+y4)/2
B. (y2+y4)
C. (y2|y4)/2
D. (y2|y4)

Question No:170 **(Marks:01)** **Vu-Topper RM**

We prefer ……over the Lagrange's interpolating method for economy of computation.

A. Newton's forward difference method
B. Newton's backward difference method
C. Newton's divided difference method
D. None of the given choices

Question No:171 **(Marks:01)** **Vu-Topper RM**

In Richardson's extrapolation method, we usually use two different step sizes ……and …… to yield a higher order method.

A. h,h/2
B. h/2,h/4
C. h/2,h/3
D. None of the given choices

Question No:172 **(Marks:01)** **Vu-Topper RM**

Simpson's rule is a numerical method that approximates the value of a definite integral by using ……polynomials.

A. Linear
B. Quadratic ok
C. Cubic
D. None of the given choices

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Question No:173 **(Marks:01)** **Vu-Topper RM**

Which of the following is the Richardson's Extrapolation limit: F3(h/8) provided that F2(h/8) = F2(h/4) = 1 ?

A. 63
B. 64
C. 1
D. -1

Question No:174 **(Marks:01)** **Vu-Topper RM**

The idea of Richardson's extrapolation is to combine two computed values of derivative of y using the same method but with …… different step sizes.

A. Four
B. Two **Page 1**
C. Three
D. None of the given choices

Question No:175 **(Marks:01)** **Vu-Topper RM**

In Romberg's method, accuracy of Simpson and Trapezoidal rules is improved by ---------.

Interpolation
Extrapolation **Google** **ok**

Question No:176 **(Marks:01)** **Vu-Topper RM**

In the process of Numerical Differentiation, we differentiate an interpolating polynomial in place of ----------.

A. actual function
B. extrapolating polynomial
C. Lagrange's polynomial
D. Newton's Divided Difference Interpolating polynomial

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Question No:177 **(Marks:01)** **Vu-Topper RM**

In Simpson's 1/3 rule, the global error is of ……

A. O(h2)
B. O(h3)
C. O(h4) **Page 171**
D. None of the given choices

Question No:178 **(Marks:01)** **Vu-Topper RM**

Which of the following is the Richardson's Extrapolation limit: F2(h/4) provided that F1(h/4) = F1(h/2) = 1 ?

A. 15
B. 16
C. -1
D. 1

Question No:179 **(Marks:01)** **Vu-Topper RM**

If f(x) = 2x, then which of the following is will be derivative of f(x) at x = 0.2?

A. 0.2
B. 0.4
C. 2
D. -2

Question No:180 **(Marks:01)** **Vu-Topper RM**

Trapezoidal and Simpson's integrations are just a linear combination of values of the given function at different values of the ……variable.

A. Dependent
B. Independent **Page 179** **ok**
C. Arbitrary
D. None of the given choices

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